

CareFlow AI

AI Tools, Capability Architecture & Usage Documentation

Document Type: Comprehensive AI & Tool Specification

Scope: Models, Capabilities, Safety Guardrails, Voice Integration, Evaluation Suite

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AI IMPLEMENTATION PRINCIPLE:

AI in CareFlow AI serves strictly as an administrative coordination engine. The AI operates within sandboxed capability interfaces (`AICapabilities`), maintaining context and resolving patient requests without direct database or clinical modification rights.

1. AI Overview & Models Used

CareFlow AI incorporates a multi-provider LLM integration layer (`app/ai/llm_client.py`) paired with a deterministic fallback NLU engine. This ensures 100% operational uptime and zero-downtime execution even during external LLM API degradation.

Provider	Model Name	Purpose	Input	Output	Code Location
Anthropic Claude	claude-3-5-sonnet-20241022	NLU intent classification & slot extraction	Message + JSON Context	Structured JSON (intent, slots)	app/ai/llm_client.py
OpenAI GPT	gpt-4o-mini	NLU intent classification & slot extraction	Message + JSON Context	Structured JSON (intent, slots)	app/ai/llm_client.py
Google Gemini	gemini-1.5-pro	NLU intent classification & slot extraction	Message + JSON Context	Structured JSON (intent, slots)	app/ai/llm_client.py
Deterministic NLU	_deterministic_nlu_fallback	Rule-assisted NLU & slot extractor (Primary Fallback)	Message + JSON Context	ExtractedNLU Pydantic Object	app/ai/llm_client.py

2. AI Runtime Intent Pipeline

At runtime, every user message passes through the NLU pipeline to extract intent and parameters into one of 10 standardized intents:

Extracted Intent	Description & Logic	Capabilities Triggered
FIND_HOSPITAL	Locate approved hospitals by city or query	search_hospitals
FIND_DOCTOR	Discover doctors by specialty or name and fetch real availability	search_doctors, check_availability
BOOK_APPOINTMENT	Select slot and execute idempotent verified booking	create_appointment, verify_external_appointment
RESCHEDULE_APPOINTMENT	Move existing appointment to new slot & release old slot	reschedule_appointment, check_availability
CANCEL_APPOINTMENT	Cancel existing booking in EHR & release slot	cancel_appointment
CHECK_APPOINTMENT	Retrieve upcoming patient bookings	get_appointment
QUESTIONNAIRE	Fetch pending pre-visit intake forms	get_questionnaire
CLARIFICATION_NEEDED	Triggered when booking request lacks doctor/specialty context	None (Prompts for clarification)
GENERAL_ADMINISTRATIVE_QUERY	Respond to general platform capabilities	None
HUMAN_ESCALATION	Triggered when emergency symptoms are detected	transfer_to_human

3. Full Inventory of Implemented AI Capabilities

All AI actions are strictly scoped to the 17 audited methods implemented in AICapabilities (`app/ai/capabilities.py`):

Capability Name	Purpose	Inputs	Outputs	Side Effects	Guardrails
1. search_hospitals	Find approved network hospitals	city, query	list of hospital objects	Database read	Audit log
2. search_doctors	Find active doctors by specialty/name	specialty, hospital_id, name	list of doctor objects	Database read	Audit log
3. check_availability	Compute real unblocked slots	doctor_id, hospital_id, date	list of available slot dicts	Database calculation	Audit log
4. create_appointment	Book slot with EHR verification	hospital_id, doctor_id, patient_id, date, start/end_time	verified appointment object	DB + EHR mutation	Idempotency + Audit
5. reschedule_appointment	Move slot & update EHR	appointment_id, new_date, new_start/end_time	updated appointment object	DB + EHR mutation	Audit log
6. cancel_appointment	Cancel booking & release slot	appointment_id, reason	cancelled appointment object	DB + EHR mutation	Audit log
7. get_questionnaire	Fetch active pre-visit form	hospital_id, appointment_id	questionnaire questions dict	Database read	Audit log
8. submit_questionnaire	Save structured intake answers	questionnaire_id, appointment_id, patient_id, responses	submission status object	Database insert + Workflow	Audit log
9. lookup_patient	Find patient profile	patient_id, email, phone	patient profile dict	Database read	Audit log
10. get_appointment	Fetch patient appointment history	appointment_id, patient_id, status	list of appointments	Database read	Audit log
11. send_notification	Dispatch user notification	recipient_id, role, type, title, message	notification record dict	Database insert	Audit log
12. start_workflow	Trigger async workflow	appointment_id, workflow_name, trigger_event	workflow execution object	Workflow engine call	Idempotency + Audit
13. get_context	Retrieve AI conversation state	conversation_id	context JSON dictionary	Database read	Audit log
14. update_preferences	Update patient comms prefs	patient_id, communication_preference	updated preference dict	Database update	Audit log
15. verify_external_appointment	Query EHR to verify record	appointment_id, external_id, idempotency_key	verification status object	EHR verification API	Audit log
16. synchronize_state	Sync internal DB with EHR	appointment_id, target_status	synchronized status dict	DB status transition	Audit log
17. transfer_to_human	Escalate to human emergency	reason, urgency	escalation dispatch object	Emergency alert trigger	Audit log

4. Voice AI Architecture & Telephony Bridge

CareFlow AI includes full voice interaction support implemented in `app/integrations/telephony.py`. It translates phone calls into streaming text turns, routes them to `run_ai_agent()`, and synthesizes audio responses.

- **Inbound Caller Identification:** Automatically matches caller phone numbers against `Patient.phone` to instantly personalize greetings and fetch context.
- **Twiml Audio Synthesis:** Generates PSTN-compatible Twiml XML payloads utilizing Amazon Polly voice (Polly.Aditi language en-IN).
- **Call Failure Handling:** Handles abnormal network drops or carrier failures gracefully, logging disconnect events and preserving conversation state for turn resumption.

5. AI Safety Guardrails & Emergency Escalation

Safety guardrails are enforced at both prompt level and code level:

Category	Enforced Boundary Rules
AI MAY DO	Discover hospitals & doctors, check real availability, book/reschedule/cancel appointments, present approved pre-visit forms, record structured responses, trigger notification workflows, escalate emergencies.
AI MUST NOT DO	Diagnose medical conditions, prescribe medications, interpret symptoms clinically, recommend medical treatment, alter clinical dosages, or invent unverified slot availability.

6. Automated AI Evaluation Framework

CareFlow AI includes an automated evaluation suite (`backend/tests/test_ai_evaluation.py`) covering 18 rigorous test cases (``Eval-1`` to ``Eval-18``). All 18 eval tests pass consistently, validating intent detection accuracy, slot extraction, context retention, clarification behavior, emergency escalations, and relative date resolution.

Test Suite Range	Evaluated Dimension	Validation Description	Result Status
Eval-1 to Eval-5	Intent Classification Accuracy	Validates FIND_DOCTOR, BOOK, CANCEL, RESCHEDULE, QUESTIONNAIRE intents	100% Pass
Eval-6 to Eval-9	Slot Extraction & Relative Dates	Validates 'tomorrow', 'Friday', specialty, and doctor name extraction	100% Pass
Eval-10 to Eval-12	Emergency Safety Escalation	Validates immediate transfer_to_human on 'chest pain' & 'severe breathlessness'	100% Pass
Eval-13 to Eval-16	Context Retention & Date Shift	Validates 'Actually make that Friday' shifting date while retaining doctor context	100% Pass
Eval-17 to Eval-18	Clarification Over Guessing	Validates CLARIFICATION_NEEDED trigger when booking request lacks doctor context	100% Pass

Document Summary

This document provides complete visibility into the AI models, tools, capabilities, voice architecture, safety guardrails, and automated evaluation metrics implemented in CareFlow AI.